

A short user manual (no more than 2000 words)

This is what you will see after opening the application:

The screenshot shows a window titled 'MainWindow' with standard window controls (minimize, maximize, close) in the top right corner. Below the title bar, there is a 'Menu:' section containing seven buttons: '1. Add a New Recipe', '2. Display Recipes', '3. View Recipe Details', '4. Scale Recipe', '5. Reset Quantities', '6. Clear All Data', and '7. Exit'. Below the menu is a 'Filter Recipes:' section with three buttons: '1. Filter by Ingredient', '2. Filter by Food Group', and '3. Filter by Max Calories'. At the bottom of the window is a large, empty rectangular area.

Adding a Recipe

- Click Add Recipe button.
- Enter the recipe name when prompted.

The screenshot shows a dialog box titled 'Input Required' with a close button (X) in the top right corner. Inside the dialog, there is a text label 'Enter recipe name:' followed by a large text input field. To the right of the input field are two buttons: 'OK' and 'Cancel'.

- Enter the number of ingredients.

The screenshot shows a dialog box titled 'Input Required' with a close button (X) in the top right corner. Inside the dialog, there is a text label 'Enter the number of ingredients:' followed by a large text input field. To the right of the input field are two buttons: 'OK' and 'Cancel'.

- For each ingredient, enter its name, quantity, calories, and food group.
- Enter the number of steps. For each step, enter a description.
- If total calories exceed 300, a warning message will display.

- Recipe added successfully message will display.

Recipe added successfully with warning: Total calories exceed 300.

Displaying Recipes

- Click Display Recipes button to view all recipes sorted alphabetically by name.

Recipes:
Recipe: cake
*** Warning: Total calories exceed 300 ***
Ingredients:
- flour: 1 (350 calories) (flour)
Steps:
1. bake the cake
Total Calories: 350

Recipe: omlett
Ingredients:
- egg: 1 (120 calories) (protein)

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Scaling a Recipe

- Click Scale Recipe button.
- Enter the number of the recipe to scale.
- Enter a scaling factor (e.g., 0.5 for half, 2 for double).
- Ingredients will be scaled accordingly, and updated recipe details will display.

Recipe: cake
Ingredients:
- flour: 2 (350 calories) (flour)
Steps:
1. bake the cake
Total Calories: 350

*** Warning: Total calories exceed 300 ***

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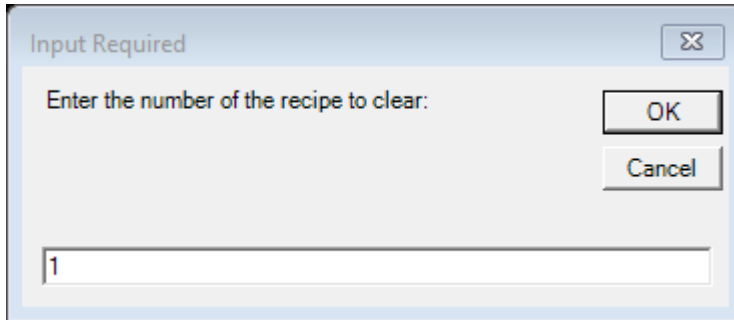
Resetting Ingredient Quantities

- Click Reset Quantities button.
- Enter the number of the recipe to reset quantities.

- Ingredient quantities will revert to their original values, and updated recipe details will display.

Clearing a Recipe

- Click Clear All Data button.
- Enter the number of the recipe to clear.
- The selected recipe will be removed from the list, and a confirmation message will display.



A screenshot of a dialog box titled "Input Required" with a close button (X) in the top right corner. The dialog contains a text prompt "Enter the number of the recipe to clear:" followed by two buttons, "OK" and "Cancel", stacked vertically. Below the buttons is a text input field containing the number "1".

Filtering Recipes

- Click Filter By Ingredient, Filter By Food Group, or Filter by Max Calories button.
- Enter the required information to filter recipes accordingly.

Filtered Recipes by Ingredient:
Recipe: omlett
Ingredients:
- egg: 1 (120 calories) (protein)
- ham: 1 (150 calories) (meat)
Steps:
1. mix and fry the ham and egg
Total Calories: 270

Exiting the Application

- Click Exit button to close the application.